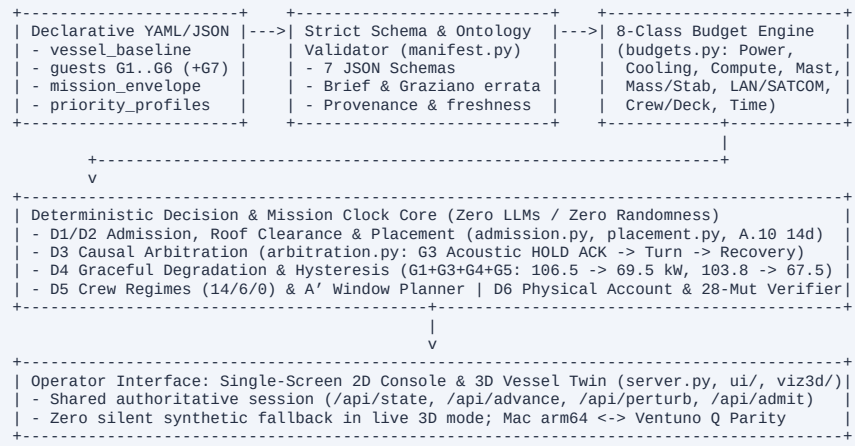
SDV Eight-Class Resource Governor (sdv-resource-governor)

Run the demo (one command): ./run_demo.sh -> opens the operator console at http://127.0.0.1:8765 (creates .venv and installs on first run).

Arduino LAN console: http://10.0.0.130:8765 runs the governor on Ventuno Q, with the browser on the Mac. This is an unauthenticated development console: use only on the trusted local network, never expose it to the Internet. Inspect or restart it with ssh arduq 'systemctl --user status sdv-governor --no-pager' or ssh arduq 'systemctl --user restart sdv-governor'.

Deterministic, ontology-driven Admission (D1), Placement (D2), Right of Way (D3), Degradation (D4), Crew as a setting (D5: 14/6/0), and Physical Account (D6), paired with a unified Operator Interface (Single-Screen 2D Console & Synchronized 3D Vessel Twin) for a 60-70 m Offshore Software-Defined Vessel (SDV).

1. Architecture Diagram & Data Flow



2. Deliverables, Objectives & Scope (D1-D6 + Operator Interface)

ID	Decision / Deliverable	Brief Tier	What it optimises (declared objective)	Component & Status
D1	Admission in 30 s (+ Guest Manifest, Annex B.0)	Mandatory	Feasibility over the whole 14-day A.10 envelope (degraded states included, never the pier); for YES_IF, displacements ranked by least capability score lost, then fewest guests displaced; states explicitly whether the verdict holds with one genset down (one_genset_down: YES / YES_WITH_STEP_DOWN / NO, plus required step-downs and binding class).	admission.py -- YES/YES_IF/NOT_NOW/NO, names the binding class (e.g. Scenario B set binds on data/SATCOM in A.10 cruise). At sea with 0 crew the verdict stays NO and adds alternative_action: LOGICAL_RECONFIGURATION with up to 3 software-only options (reconfiguration.py; apply via POST /api/apply_logical_reconfig). G1-G6 cards + own G7 card, schemas/*.schema.json.
D2	Placement	One of D2/D4 (addressed)	Objective D2_LEXICOGRAPHIC_V2, key (roof_cond_rank, pref_penalty, -retention_milli, scarcity_cost, -min_margin_bucket, stab_eq, node_used_cores, nid): avoid conditional roof positions -> preferred positions -> capability retained after environmental effects -> least scarce slots (free mast = 3) -> largest minimum margin over the budget classes -> minimum stability-equivalent mass -> least-loaded compute node -> lexicographic node; up to 5 ranked feasible placement_alternatives with why_not_chosen and up to 5 rejected_placements with their binding witness.	placement.py -- bay/mast/deck/power line/cooling manifold; ROOF_OUTSIDE_G4_RECOVERY_V1 conditional roof clearance, enforced in D1/D3 (scheduling.py::evaluate_roof_drone_clearance_safe).
D3	Right of way in 2 s (Scenario B min 8)	Mandatory	Max utility_score = priority-weighted survey continuity + priority-weighted drone landing margin, subject to the hard G4 <20 % battery deadline; tie-break declared in the priority file.	arbitration.py -- one-sentence captain rationale + cost to the loser; survey_primary lands G4 at T+1095 s (105 s margin, G3 loses 14 min), rescue_primary at T+855 s (345 s margin, G3 loses 26 min).
D4	Degradation (Scenario B min 12)	One of D2/D4 (addressed)	Restore power/cooling feasibility with least capability lost: step down before shedding, in the priority file's declared order; never shed safety-critical, uninterruptible, airborne or submerged guests.	degradation.py -- generator trip at T=720 s: committed 106.5 kW > 100 kW -> G5 45 -> 8 kW; rationale lists what every guest keeps or loses.
D5	Crew as a setting (Scenario C)	Design deliverable	Same rule, crew-dependent timings: 14/6/0 aboard -> 30 s/120 s/automated confirmation, 240 s/600 s/prohibited deck clearing.	scheduling.py, docs/MEMO_SCENARIO_C.pdf.
D6	Physical Account (Annex C)	Required deliverable	Every assumed number traceable to the code line that uses it.	docs/PHYSICAL_ACCOUNT_D6.pdf, ASSUMPTIONS.yaml, fail-closed evidence verifier (28 mutations, false_accepts=0).
UI	Operator Interface (Single-Screen 2D Console & 3D Vessel Twin)	Operator Interface	One screen showing the same authoritative session the engine decides on.	Implemented: Zero-build 2D console (http://127.0.0.1:8765) + Three.js 3D twin (viz3d/) bound to the same authoritative GovernorSession (visible STALE / DESCONECTADO banner if disconnected).

3. Clean / Offline Installation & Quickstart (<= 5 minutes)

```
# 1. Create clean virtual environment (offline wheelhouse or system site-packages supported)
python3 -m venv .venv --system-site-packages
.venv/bin/pip install -r requirements-macos-arm64.lock
.venv/bin/pip install --no-build-isolation --no-deps -e .

# 2. Run the full automated test suite (Mac arm64 or Linux aarch64)
.venv/bin/pytest -v

# 3. Generate deterministic campaign bundle & run fail-closed verifier + 28 adversarial mutations
.venv/bin/sdv-governor campaign --run-id demo_run --junit-xml artifacts/final_run/junit.xml
.venv/bin/sdv-governor verify-evidence artifacts/demo_run --run-mutations

# 4. Launch the interactive single-screen 2D Operator Console (Ctrl+C to stop)
.venv/bin/sdv-governor serve --host 127.0.0.1 --port 8765
```

4. Eight-Class Resource Model (PHY-01..PHY-10)

Class	Physical / Operational Constraint	Source Module
1. Power	200 kW net reserve (LINE_A_FWD <= 150 kW, LINE_B_AFT <= 150 kW), actual vs. committed firm reservations (106.5 kW -> 69.5 kW)	budgets.py

2. Cooling	Scenario B nominal: 24 deg C, 133.3 kW water capacity; generator trip at T=720 s is an electrical fault, not a cooling-loop loss. Separate hot-water stress case: 30 deg C, 120 kW capacity (72 kW with -40% loop loss; committed heat 103.8 -> 67.5 kW). Thermal inertia: 30 kW x 60 s (1.8 MJ).	budgets.py
3. Compute	4 physical rack nodes (node_1..node_4, 32c / 128 GiB / 1 acc each; 80% allocatable = 25c / 102.4 GiB per node, 3 assignable discrete accelerators across cluster, 16 TB storage)	placement.py
4. Mast & Roof	mast_3 (<= 60 kg, <= 0.6 m2, <= 2,000 W air) + hangar_roof (<= 150 kg, <= 2.0 m2, <= 2,000 W air, conditional ROOF_OUTSIDE_G4_RECOVERY_V1)	budgets.py
5. Mass & Stability	Global <= 10,000 kg and <= 12,000 kg-eq (G1+G3+G4+G5 = 9,560 kg, 9,853 kg-eq, G3+G4+G5 = 9,515 kg ~= 9.52 t); bay1 <= 8t, bay2 <= 4t	budgets.py
6. Data Links	10 Gbps LAN (<= 1 Gbps/port), SATCOM 8 Mbit/s nominal (brief A.5: shared 2-8 Mbit/s) / 6 Mbit/s A.10 cruise / 2 Mbit/s degraded / 0 bps EMCON (W4)	budgets.py
7. Crew & Deck	Regimes 14 (9 deck avail, 30 s confirm, 240 s deck clear), 6 (2 deck avail, 120 s confirm, 600 s deck clear), 0 (0 avail, manual evolutions prohibited)	scheduling.py
8. Time	1 Hz tick budget (<= 1,000 ms), Scenario A admission (<= 30 s), A.6 transition durations, local urgent command (<= 200 ms, < 1 ms measured)	replay.py

5. Adding a New Guest (G7) Without Touching Code (EXT-01)

Any new guest payload is admitted purely from a declarative dictionary or YAML file:

```
import yaml
from sdv_governor.manifest import parse_guest_manifest

raw = yaml.safe_load("""
schema_version: "sdv-guest-manifest/1.0"
guest_id: "G7"
name: "G7 - Compact Expendable Bathythermograph (XBT) Launcher"
category: "AUXILIARY_PAYLOAD"
safety_critical: false
default_mode: "STANDBY"
can_buffer_offline: true
subcomponents:
  - {subcomponent_id: "G7_DECK_LAUNCHER", part_of: "G7", name: "Stern XBT Pod", mass_kg: 180,
     footprint_m2: 1.2, wind_area_m2: 0.8, preferred_positions: ["bay1_b", "bay2"],
     allowed_positions: ["bay1_b", "bay2", "bay1_a"], cooling_route: "CHILLED_WATER_LOOP",
     requires_water_manifold: false, value_origin: "ASSUMED"}
modes:
  STANDBY:
    {description: "Launcher indexed and heater idle", min_power_w: 200, nominal_power_w: 300,
     sustained_power_w: 300, sustained_duration_s: 3600, peak_power_w: 500, peak_duration_s: 10,
     cooling_water_w: 300, cooling_air_mast_w: 0, compute_cores: 1, compute_ram_mib: 1024,
     compute_accelerators: 0, shared_storage_gb: 20, backbone_lan_bps: 1000000, satcom_bps: 0,
     safety_critical: false, uninterruptible: false, min_done_duration_s: 0, min_cooldown_s: 0,
     forbids_emcon_silence: false, requires_deck_clear: false, requires_quiet_acoustic_state: false,
     capability_retention_ratio: 0.25, value_origin: "ASSUMED"}
installation: {port_only: true, min_crew_required: 2, install_time_min: 30, remove_time_min: 20,
               sea_install_max_wind_kn: 0, sea_install_min_speed_kn: 0.0, sea_install_max_speed_kn: 0.0}
vehicle_replacement: {supported_at_sea: false, min_crew_required: 0, swap_duration_min: 0,
                      min_speed_kn: 0.0, max_speed_kn: 0.0, max_wind_kn: 0, max_sea_state: 0}
ship_state_constraints: {min_speed_kn_for_operation: 0.0, max_speed_kn_for_operation: 20.0,
  max_wind_kn_for_operation: 35, max_sea_state_for_operation: 6,
  requires_heading_into_wind_for_launch_recovery: false, forbids_deck_machinery_while_active: false,
  acoustic_hold_handshake_min_s: 0, acoustic_hold_handshake_max_s: 0, acoustic_hold_default_s: 0,
  low_battery_threshold_pct: null, emergency_landing_deadline_s: null, max_flight_endurance_min: null}
dependencies: {requires_guests: [], excludes_guests: [], conditional_excludes: []}
mission_capabilities: {hydro_sampling_value: 20.0}
""")
manifest = parse_guest_manifest(raw)
```

If the new card launches or recovers a craft, set ship_state_constraints.launch_recovery_area to flight_deck or stern_ramp (default none). The flight deck <-> stern ramp exclusion (Annex A.3) is then inherited from the mutually_exclusive_with_* flags of data/ship/vessel_baseline.yaml (areas.py); asymmetric flags are rejected at load time (fail-closed).

If a mode needs the acoustic command link (G3 HOLD_ACOUSTIC), set requires_acoustic_command_link: true on it. The link (acoustic.py, data/ship/acoustic_link_policy.yaml) is down under layered water (water_column_layered, scenario/operator input, ASSUMED), own-ship manoeuvre or deck machinery; the governor then plans G3 to hold autonomously after the 120 s lost-link timeout (ASSUMED) instead of assuming a 15 s ACK.

6. Explicit Limitations & Honest Operational Boundaries

* 1. Software Dispatch vs. Physical Ship Actuation (< 1 ms vs. Physical Plant): The < 1 ms (~0.015 ms on Mac arm64, ~0.032 ms on Ventuno Q aarch64) latency measured for URGENT_STOP / HOLD / SECURE reflects local Python governor state transition and command emission only (<= 200 ms software budget). Real vessel breaker opening, hydraulic winch braking, acoustic modem propagation (15 s nominal delay modeled in D3), and vessel heading stabilization (90 s modeled in D3) are physical plant processes governed by external PLC/PMS hardware.

* 2. Discrete Simulation Resolution & Linearized Plant Curves: Scenario B executes at 20 s discrete ticks (90 intervals / 91 ticks over 1800 s, with 1 Hz internal causal verification) with piecewise-linear seawater cooling capacity (160 kW -> 120 kW) and a single-node lumped thermal inertia reservoir (1.8 MJ, 600 s cooldown) rather than a continuous CFD/pipe-network solver.

* 3. Live 90-Second Human Rehearsal Status (DEFERRED_OWNER): While the 90-second (20x compressed) Scenario B playback, API runner, and operator script (docs/DEMO_SCRIPT_90S.md) are fully implemented and verified by automated tests, the live human spoken rehearsal with the project owner is explicitly recorded as DEFERRED_OWNER.

* 4. Hardware Peripheral Scope: STM32H5 MCU real-time motor control is not required by the naval governor challenge; Ventuno Q (training06, aarch64) executes the full Python governor and verification suite over LAN (ssh ardu).

7. What We Would Do With One More Week

* 1. Closed-Loop HIL Bus Integration (Modbus TCP / NMEA 2000 / CAN): Connect FeedReading telemetry and pending_commands directly to a hardware-in-the-loop PMS/winch simulator over NMEA 2000 and Modbus TCP with hardware watchdog heartbeats on the Ventuno Q STM32H5 co-processor.

* 2. Continuous Multi-Compartment Thermal & Power Transient Model: Replace single-reservoir thermal inertia with per-bay chilled-water flow/temperature differential equations and generator governor droop curves during 35 kW UAV arrestor peaks.

* 3. Rolling-Horizon Weather & Sea-State Forecast Arbitrator: Extend D3/D5 arbitration from a single 30-minute conflict window to a 72-hour rolling metocean forecast that pre-positions UAV recovery and battery charging before sea-state degradation.

* 4. Cryptographic Signed Audit Ledger: Sign every evidence_manifest.json and live operator override (URGENT_STOP, apply_yes_if) with hardware-backed Ed25519 keys for maritime regulatory black-box compliance.

8. Public Data (Golden Gate) & Consumption Forecasting (Advisory)

Integrates offline recorded coastal metocean series at Golden Gate Bar (GG_BAR: lat 37.788, lon -122.634) with an advisory 1 Hz damped Holt-Winters guest power forecaster, split-conformal confidence bands, and exact physical budget projection.

8.1 What It Does & Operational Workflow

* Public Metocean Replay & Advisory Outlook: Injects a recorded (offline) snapshot of SST, wave height (Hs), sea state (WMO 3700) and wind into replays, and computes guarded, fail-closed operational windows over a 48 h outlook; when none exists it states why (weather or operational blockers).

* Predictive Advisory Forecaster (E2): Recursively estimates normalized power level and damped trend (phi = 0.98) at 1 Hz, projecting conformal bounds across h in {10, 20, 30, 45, 60} s. When projected upper bounds exceed 110% of declared load (ASM-FCST-OVERRUN-THRESHOLD), a bisection search over the 8-class budget engine identifies the binding bottleneck (<= 8 evaluations; p99 full tick 2.68 ms on Ventuno Q) and issues advisory alerts before reactive shedding (benchmark median lead 60.5 s; Golden Gate replay ~37 s). /api/forecast is a labelled REPLAY with SIMULATED telemetry.

* Strict Non-Interference: The forecaster runs outside the authoritative decision core (observer=None by default) and never modifies governor verdicts, priority shedding, or semantic hashes (FC-INT-01..03).

8.2 CLI & Quickstart

```
# Validate offline metocean series and run Golden Gate environmental scenarios
.venv/bin/sdv-governor public-env-suite

# Benchmark damped Holt forecaster against pre-registered acceptance gates
.venv/bin/python tools/bench_forecast.py --split both   # or: sdv-governor forecast-bench
```

8.3 Pre-Registered Acceptance Gates & Validation Results

Measured on the production observer (docs/evidence/forecast_report_20261018.md); thresholds unchanged.

Gate	Requirement	Measured	Status
G-NI	Campaign hash/decisions unchanged with observer	6 tests; 3a0aa7f6? (scope fixes C2-C6; artifacts/merge_scope/GOLDEN_DIFF.md)	PASS
G-OFF	Offline runtime, zero network	2 tests	PASS
G-LAT	p99 observer tick <= 50 ms	2.68 ms on Ventuno Q (0.91 ms Mac)	PASS
G-ACC-RAMP / STEADY	ratio MAE@30 <= 0.85 / <= 1.05	0.785 / 0.858	PASS
G-COV	90% band coverage in [0.85, 0.95]	0.908	PASS
G-RECALL / G-LEAD	recall >= 0.90; lead median >= 20 s, p10 >= 10 s	1.00; 60.5 s / 37.9 s	PASS
G-FA	<= 1.0 false WARN / h	0.115 / h	PASS
G-OOD	OOD recall >= 0.75 and FA <= 3 / h	recall 0.90, FA 1.5 / h (OOD guard; was 9.7)	PASS
G-ENV	0 false-open windows with guard (7-day backtest)	0 guarded vs 20 raw	PASS

8.4 Honest Operational Disclosures & Verification Boundaries

* "Recorded public data snapshot (UTC 2026-10); runtime is offline."
* "Open-Meteo is a coastal model product at ~8 km; validated against NDBC 46237/46026/FTPC1; derived guard bands +7.3 kn wind, +0.24 m Hs, +0.51 deg C SST."
* "On unfamiliar telemetry an OOD guard holds WARN to point-forecast risk sustained 6 s (G-OOD: 1.5 false WARN / h, recall 0.90); it is advisory only."
* "Consumption telemetry is synthetic (seeded); the forecaster is advisory and never changes the governor’s decision."
* "At Golden Gate SST (~16 deg C), cooling is not the binding constraint; 14-day admission is still judged at 30 deg C."

9. Repository Structure

* src/sdv_governor/ -- Deterministic engine (models.py, manifest.py, budgets.py, placement.py, admission.py, arbitration.py, degradation.py, scheduling.py, replay.py, evidence.py, server.py, cli.py, pdf_export.py).

* data/ & schemas/ -- Vessel baseline (vessel_baseline.yaml, mission_envelope_a10.yaml), guest manifests (G1-G6), priority profiles, scenarios, errata/clarifications, and 7 JSON Schemas (mirrored in src/sdv_governor/_bundled/).

* ui/ & viz3d/ -- Single-screen 2D Operator Console (ui/) and synchronized Three.js 3D Vessel Twin (viz3d/).

* docs/ -- Deliverables (PHYSICAL_ACCOUNT.md / PHYSICAL_ACCOUNT_D6.md, MEMO_SCENARIO_C.md / SCENARIO_C_MEMO.md, PITCH_SLIDES.md / SLIDES_10.md, DEMO_SCRIPT_90S.md, DEPLOYMENT_AND_BOM.md + PDFs).

* tests/ & artifacts/ -- Unit, integration, and adversarial regression suites + verified release bundles (release_s6, release_s6_patch1, release_alignment_graziano_v1).